this is approaching the limit using oil immersion with a 50-cycle/sec. frequency. As Gregory and Donaldson mention, without correction the image in the central region of the scan would be weak, while that at the limits would be very bright by comparison. In the present apparatus the image brightness is rendered uniform by changing the phase so that the lamp brightness is proportional to the velocity of the scanning plane. By suitable adjustment of the impedance of the vibrator supply circuit, this supply is displaced some 80° out of phase with the supply to the microscope lamp. Thus, allowing for the lag in the filament temperature, the brightness increases with the velocity of the scan and falls to a minimum at the end of the stroke where the velocity is momentarily zero. Further critical control can then be exercised by varying the current to the lamp, thus altering the hysteresis in the filament.

The apparatus can be adjusted to give surprisingly good resolution at magnifications of × 1,000, using oil immersion objectives, provided that amplitudes less than about 15μ are used. No trouble was experienced with non-drying immersion oils of normal viscosity. The main difficulty at present is that standing waves and other imperfections leading to non-linear motion are present in the vibrator diaphragm. These may be corrected for any position; but readjustment is required with each new position of the slide. The inherent problem is to provide strictly linear motion without increasing bulk or momentum.

W. D. JACKSON

Botany Department,
University of Tasmania,
Hobart. Jan. 28.

[1] Gregory, R. L., and Donaldson, P. E. K., *Nature*, **182**, 1434 (1958).

GEOPHYSICS and GEOCHEMISTRY

Remanent Magnetism of the Cretaceous Basalts of Madagascar

A GENERAL study of the remanent magnetism of volcanic rocks from Madagascar has been undertaken since the beginning of 1958 under the direction of Dr. H. Besairie, chief of the Geological Department of Madagascar.

From various places in the island, more than a hundred oriented samples have been collected by one of us (L. C.) and seven samples by J. Boulanger and P. Bussiere. Forty samples belong to flows or dykes well dated from the upper Cretaceous. Their ages have been proved to vary between Cenomanian and Santonian[1].

The samples were taken in places widely distributed along the eastern coastal district and the western part of the island. They were generally collected in quarries or trenches, some metres below ground-level, but sometimes at ground-level in river beds. The rocks are dolerites, basalts or sakalavites.

In some places, Ampijoroa, Amborompotsy, Antsalova (part), the directions, as well as the values of the intensities of magnetization, proved widely scattered. Values as high as 33×10^{-3} c.g.s. units/gm. were found. It is believed[2,3] that lightning currents were responsible for these magnetizations. For the other places the intensities as well as the directions of magnetizations showed good consistency.

For samples originated from the same flow or dyke, the differences between the individual values obtained and the mean value vary from 0 to 4 deg. for inclination and from 0 to 20 deg. for declination.

According to the places, the mean values found for the intensities of magnetization vary from $0{\cdot}8 \times 10^{-3}$ to 9×10^{-3} c.g.s. units/gm.

The magnetic stability of the samples was tested by E. Thellier's method[4], that is, turning over the sample in the Earth's magnetic field for one to several months and comparing the values obtained before and after turning over. Except for two samples, the magnetization showed no variation.

The nature of the magnetization was tested on some samples by successive heatings in a non-magnetic furnace. Only one Curie point was observed between 400° C. and 600° C. for each sample. The magnetization is thought to be thermoremanent[5] and acquired at the time of cooling of the rocks.

All the samples are magnetized according to the sense of the present geomagnetic field at Madagascar. The mean values of the directions of magnetization for the stable samples at each place are given in Table 1. The inclinations are termed negative when the extremity of the magnetization vector points above the horizon.

Table 1

	Inc.	Dec.
Northern districts		
Sambava	−49° 30′	15° E.
Antalaha	−55°	39° E.
Samanavaka	−48°	18° W.
Central districts		
Fenerive	−62°	19° E.
Andramy	−51°	7° E.
Andafiha	−57°	18° E.
Antsalova	−47°	4° E.
Southern districts		
Manakara	−70°	30° E.
Vineta	−53°	26° E.
Tongobory	−56°	15° E.

The inclinations are systematically steeper by 2–10 deg. than those of the present magnetic field for the places on the eastern coast (Sambava, Antalaha, Fenerive, Manakara), but generally lower for the western districts (the differences from the present magnetic field being less than 7 deg.). Notwithstanding a dispersion of some degrees, the magnetic inclination of the samples increases with increasing latitude.

On the whole there is not much difference between the distribution of the magnetic inclination of the samples and that of the present geomagnetic field in the island, as we noticed in our first studies.

The declinations of the samples are all easterly, except for one place (Samanavaka). They differ widely from those of the present geomagnetic field in the island (which vary from 7° W. to 19° W.), the mean difference being nearly 30 deg.

The calculation of the orientation of the dipole axis corresponding to the magnetization of the rocks gave the values for the co-ordinates of the north geomagnetic pole as shown in Table 2.

Table 2

	Lat.	Long.
Northern districts	70° N.	161° W.
Central districts	70° N.	165° W.
Southern districts	64° N.	177° W.
Mean	68° N.	168° W.

The place calculated is in the vicinity of the Behring Strait (Fig. 1). S. K. Runcorn[7] has given results he obtained from the Cretaceous Dakota Sandstones the magnetization of which could correspond to a north geomagnetic pole at latitude 77° N. longitude 127° E., the sense of magnetization being

©1959 Nature Publishing Group

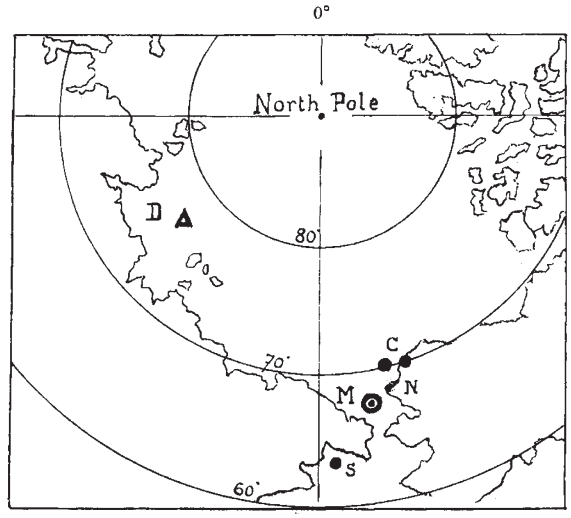


Fig. 1. North geomagnetic poles for Dakota (D, △); Madagascar, mean (M, ○); Madagascar, parts (N, C, S, ●)

reversed compared with that of the present terrestrial magnetic field.

The north geomagnetic pole for Madagascar is thus at a distance of less than 1,500 miles from the north geomagnetic American pole as given by Runcorn, and a little farther from the present geographical north pole.

This can be interpreted in different ways.

(1) The geomagnetic field, though having a dipole character for the main part, may have presented at times in the past a more or less complicated pattern, and the anomalies were smoothed out by taking the means of values scattered over long periods (of perhaps thousands or millions of years).

Therefore our results would have no significance in the perspective of polar wandering or continental drift.

(2) According to Runcorn's hypothesis[8], the mean geomagnetic field obtained by taking the mean of values scattered over a long period is a geocentric dipole field; our results would give the mean orientation of the dipole relative to Madagascar during a part of the upper Cretaceous.

(*a*) If the axis of the dipole had undergone, in the past, angular displacements of limited amplitude comparatively rapidly, say, for example, 10–20 deg. in one or a few million years, then the difference between our results and those of Runcorn could be most simply interpreted by the fact that the samples of America and of Madagascar do not belong exactly to the same period and the results give a picture of the orientation of the dipole at two different periods of the Cretaceous.

The opposed polarities of magnetization for the samples of Dakota and Madagascar give support to this hypothesis.

(*b*) If the dipole axis has undergone in the past displacements only at a slow rate (whether the dipole coincided or not with the axis of rotation of the Earth), we would have to think of relative continental drift for Madagascar and North America. From the Cretaceous up to the present time, according to calculation, the drift for Madagascar would have been about 1,500 miles in amplitude with a direction north-north-easterly.

Though displacements of much larger amplitude have been contemplated for Australia[9] and the Deccan peninsula[10], this hypothesis is considered by us as less likely than that of paragraph 2*a* above.

We wish to express our thanks to Prof. E. Thellier, whose advice was of inestimable value, and to Dr. H. Besairie, who has organized the sampling part of the work and provided the stratigraphical dating. We are also indebted to Prof. H. Dessens, director of the Puy de Dôme Observatory, and to Prof. M. Roques, who have greatly facilitated these studies.

A. Roche

Observatoire du Puy de Dôme,
Université de Clermont,
12 Avenue des Landais,
Clermont-Ferrand.

L. Cattala

Observatoire de Tananarive.

Feb. 10.

[1] Besairie, H., and Collignon, M., "Lexique internat. de Stratigraphie", **4**, Afrique, 2, Madagascar, 24 (1956).
[2] Roche, A., thesis, Paris (1953).
[3] Hales, A. L., *J. Mon. Not. Roy. Astro. Soc.*, Geophys. Supp., **1**, 3, 249 (1958).
[4] Thellier, E., *C.R. Acad. Sci., Paris*, **204**, 876 (1937).
[5] Thellier, E., *J. de Phys.*, **12**, 205 (1951).
[6] Roche, A., Cattala, L., and Boulanger, P., *C.R. Acad. Sci., Paris*, **246**, 2922 (1958).
[7] Runcorn, S. K., *Bull. Geol. Soc. Amer.*, **67**, 301 (1956).
[8] Creer, K. M., Irving, E., Runcorn, S. K., *Phil. Trans. Roy. Soc.*, A, **974**, 250, 144 (1957).
[9] Irving, E., *Mon. Not. Roy. Astro. Soc.*, Geophys. Supp., **1**, 3, 224 (1958).
[10] Clegg, J. A., Radakrishnamurty, C., and Sahasrabudhe, P. W., *Nature*, **181**, 830 (1958).

Biological Origin of Manganese-rich Deposits of the Sea Floor

Recently, one of us has advanced arguments[1] in support of the idea that manganese-rich deposits of the sea floor may result from the metabolic activity of a bottom-dwelling organism that derives its nutrients from organic complexes in the water. We present here some new evidence that is relevant to that concept.

On December 18, 1958, at lat. 34° 48′ N., long. 60° 30′ W., in 5,139 m. of water, Mr. L. V. Worthington on *Chain* Cruise No. 2 retrieved about 50 gm. of light tan mud and preserved it for us in borate-buffered (*p*H 8) formalin. The present bottom temperature at this site is 2·3° C. The finer fractions of this material were removed by suspension in water and decantation, and it was possible to isolate a few mgm. of a relatively coarse fraction, which, for the most part, consists of the vacant tests of agglutinated arenaceous (? *Rhabdammina* sp.) and of calcareous Foraminifera. It is of particular interest that almost all the former are coated inside and out with a dark brown or black, earthy or shiny veneer which, by one-dimensional paper chromatography, is seen to be markedly similar in composition to manganese-rich nodules from the Blake plateau, that is, containing appreciable quantities of nickel, cobalt and copper as well as abundant manganese and iron. Treatment with 6 *N* hydrochloric acid removes the dark coating and leaves the Foraminifera tests intact, thus demonstrating that the manganese-rich material was formed as a surface deposit, and not through the capture of pre-existing manganese-rich particles by the living Foraminifera. All manganese-rich particles in this sample apparently were formed by deposition on agglutinated Foraminifera the natural surfaces of which seem to be studded with detrital quartz grains; no comparable manganese-rich deposits are visible

©1959 Nature Publishing Group